



Solution Set

Question 1

A two-asset mean-variance investor maximizes

$$\mu^\top w - \lambda w^\top \Sigma w$$

subject to

$$\mathbf{1}^\top w = 1$$

Derive the first-order condition and solve for w in terms of the Lagrange multiplier.

Answer key

Use the Lagrangian

$$\mathcal{L}(w, \eta) = \mu^\top w - \lambda w^\top \Sigma w - \eta(\mathbf{1}^\top w - 1)$$

The stationarity condition is

$$\mu - 2\lambda \Sigma w - \eta \mathbf{1} = 0$$

so

$$w = \frac{1}{2\lambda} \Sigma^{-1} (\mu - \eta \mathbf{1})$$

The multiplier η is then chosen so that $\mathbf{1}^\top w = 1$.

Question 2

For

$$\Sigma = \begin{bmatrix} 0.04 & 0.018 \\ 0.018 & 0.09 \end{bmatrix}, \quad \mu = (0.06, 0.10)^\top, \quad \lambda = 2$$

compute the unconstrained mean-variance stationary portfolio with

$$\mathbf{1}^\top w = 1$$

Answer key

Use the formula from Question 1. Numerically,

$$\Sigma^{-1} \approx \begin{bmatrix} 30.864 & -6.173 \\ -6.173 & 13.717 \end{bmatrix}$$

Solve for η using $\mathbf{1}^\top w = 1$:

$$w = \frac{1}{4} \Sigma^{-1} (\mu - \eta \mathbf{1})$$
$$\eta \approx 0.0232, \quad w \approx (0.982, 0.018)$$

The high risk aversion and larger second-asset variance push almost all weight into asset 1 despite asset 2 having higher expected return.

Question 3

Prove that

$$w^\top \Sigma w$$

is convex when

$$\Sigma$$

is positive semidefinite.

Answer key

For $f(w) = w^\top \Sigma w$ with symmetric Σ , the Hessian is

$$\nabla^2 f(w) = 2\Sigma$$

A twice differentiable function is convex when its Hessian is positive semidefinite everywhere. Since Σ is positive semidefinite, 2Σ is also positive semidefinite. Hence portfolio variance is a convex function of weights.

Question 4

Minimize

$$f(x, y) = x^2 + 4xy + 5y^2 - 2x$$

under no constraints. Classify the stationary point using the Hessian.

Answer key

The gradient is

$$\nabla f = (2x + 4y - 2, 4x + 10y)$$

Set it to zero:

$$\begin{aligned} 2x + 4y &= 2, & 4x + 10y &= 0 \\ y &= -2, & x &= 5 \end{aligned}$$

The Hessian is

$$H = \begin{bmatrix} 2 & 4 \\ 4 & 10 \end{bmatrix}$$

The leading principal minors are $2 > 0$ and $20 - 16 = 4 > 0$, so H is positive definite. The stationary point is a strict global minimum.

Question 5

State the KKT conditions for minimizing

$$f(x)$$

subject to

$$g_i(x) \leq 0, \quad h_j(x) = 0$$

Explain each condition in finance language.

Answer key

The KKT conditions are

$$\begin{aligned} \nabla f(x^*) + \sum_i \lambda_i \nabla g_i(x^*) + \sum_j \nu_j \nabla h_j(x^*) &= 0 \\ g_i(x^*) &\leq 0, \quad h_j(x^*) = 0, \quad \lambda_i \geq 0, \quad \lambda_i g_i(x^*) = 0 \end{aligned}$$

Stationarity balances marginal objective improvement against marginal constraint costs. Feasibility means the portfolio obeys limits. Dual feasibility means tightening a limit cannot reduce minimum cost. Complementary slackness says only binding limits have nonzero marginal value.

Question 6

A trader maximizes log utility

$$\mathbb{E}[\log(1 + wR)]$$

for a small risky allocation w . Use a second-order approximation to derive the approximate optimal w .

Answer key

For small wR , use

$$\log(1 + wR) \approx wR - \frac{1}{2}w^2R^2$$

Taking expectations gives

$$\mathbb{E}[\log(1 + wR)] \approx w\mathbb{E}[R] - \frac{1}{2}w^2\mathbb{E}[R^2]$$

The first-order condition is

$$\mathbb{E}[R] - w\mathbb{E}[R^2] = 0$$

so

$$w^* \approx \frac{\mathbb{E}[R]}{\mathbb{E}[R^2]}$$

This is an interview classic because it connects Taylor expansion, optimization, and risk sizing.

Question 7

A nonlinear optimizer returns a solution at a VaR limit boundary. What checks establish whether the boundary solution is economically credible?

Answer key

Check KKT stationarity, whether the VaR constraint is smooth enough near the solution, whether the VaR estimate is stable under resampling, whether finite-difference gradients agree with analytical gradients, and whether small changes in inputs cause large changes in positions. Boundary optima are common, but fragile boundary optima require skepticism.

Question 8

Explain the difference between local and global minima, and give a calibration example where this matters.

Answer key

A local minimum is best only within a neighborhood; a global minimum is best over the full feasible set. In volatility-surface calibration, different parameter sets can fit different regions of the surface. A local optimizer can settle in a basin that fits ATM options but misses wings, so interviewers expect multi-start, parameter bounds, and residual diagnostics.

Question 9

For $f(x,y)=(x-1)^4+(y+2)^2$, explain why the Hessian test is inconclusive at part of the optimum but the point is still a minimum.

Answer key

The gradient vanishes at $(1,-2)$. The Hessian is

$$H = \begin{bmatrix} 12(x-1)^2 & 0 \\ 0 & 2 \end{bmatrix}$$

At $(1,-2)$,

$$H = \begin{bmatrix} 0 & 0 \\ 0 & 2 \end{bmatrix}$$

which is positive semidefinite, not positive definite. The usual strict second-order test is inconclusive. But each term in f is nonnegative and both are minimized at $(1,-2)$, so the point is a global minimum.

Question 10

Derive the minimum-variance hedge ratio for minimizing

$$\text{Var}(P - hF)$$

Answer key

The objective is

$$\text{Var}(P - hF) = \text{Var}(P) - 2h\text{Cov}(P, F) + h^2\text{Var}(F)$$

Differentiate with respect to h :

$$-2\text{Cov}(P, F) + 2h\text{Var}(F) = 0$$

Hence

$$h^* = \frac{\text{Cov}(P, F)}{\text{Var}(F)}$$

This is the same slope one obtains from regressing portfolio PnL on futures PnL.

Question 11

What does the Lagrange multiplier on a portfolio budget constraint mean?

Answer key

The multiplier measures how much the optimal objective changes when the budget is relaxed by one unit. In a maximization problem, it is the marginal value of capital. In a minimization problem, it is the marginal cost reduction from allowing more budget.

Question 12

Why do quant interviewers ask whether the covariance matrix is positive semidefinite before discussing optimization?

Answer key

If the covariance matrix is not positive semidefinite, some portfolio can have negative computed variance, which is impossible. The optimization can become nonconvex or economically meaningless. A rigorous answer is to check eigenvalues and, if needed, repair the matrix by cleaning correlations, shrinking estimates, or projecting to the nearest positive semidefinite matrix.